Document Lens

User Manual

Batch PDF Analysis for Research & Compliance

Version 0.21.0

June 2026

Document Lens Team

Contents

Getting Started

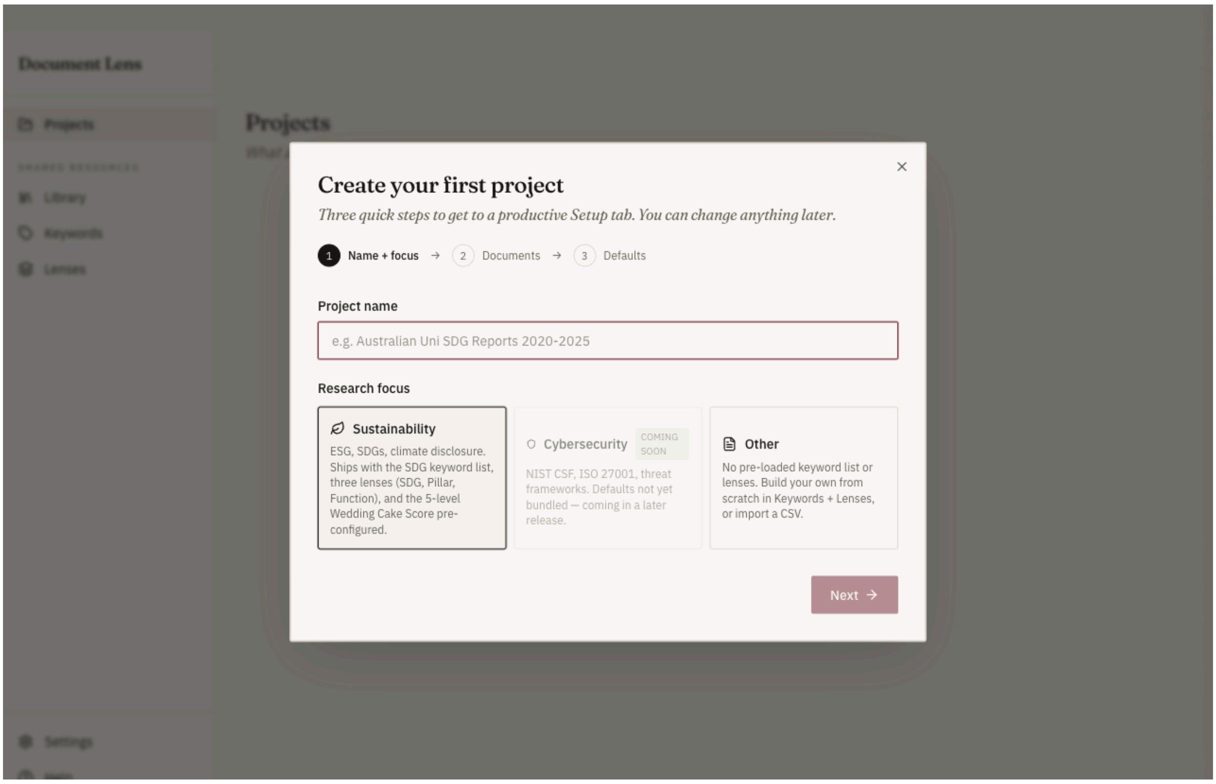

Figure 1: The three-step first-run wizard — name, focus, documents, defaults.

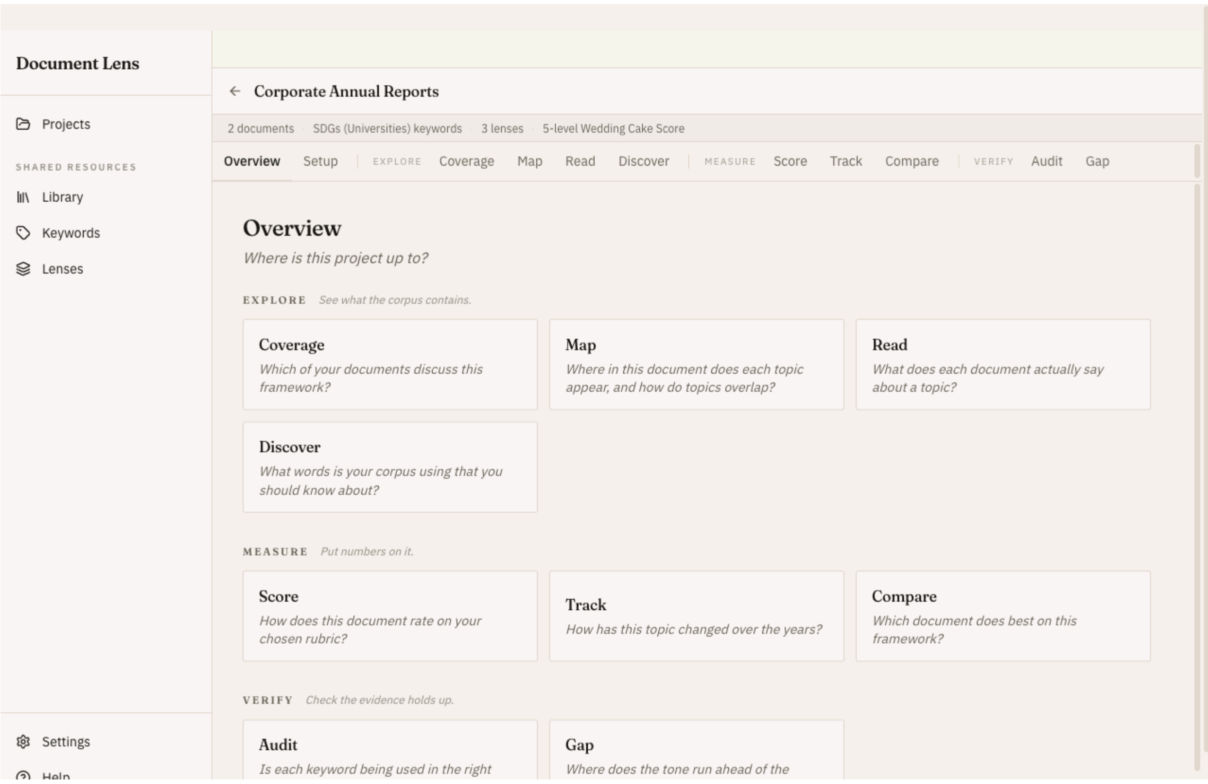

Figure 2: The project Overview: every workflow as the question it answers.

Document Lens is a workspace for keyword analysis of document corpora. The intended workflow: assemble a **project**, then work through the workflows in three phases — **Explore** what the corpus contains (Coverage, Map, Read, Discover), **Measure** it (Score, Track, Compare), and **Verify** the evidence holds up (Audit, Gap). Each workflow answers one research question; the project's **Overview** tab lists them all as those questions, so you can navigate by what you want to know rather than by tool name.

Your first project

From the empty **Projects** page, click *Create your first project* to launch the three-step wizard. Defaults are pre-loaded for sustainability research: the SDG keyword list, three lenses (SDG, Pillar, Function), and the 5-level Wedding Cake Score. Pick documents (or skip and add them from the Library later), confirm the defaults, and you'll land on the project's **Setup** tab with everything wired up. After that, reopening a project resumes the workflow you last had open.

The data model in one paragraph

Each **keyword** carries a **polarity**: positive (signals delivery — "carbon reduction") or counter (signals greenwashing — "performative disclosure"). Keywords carry tags against keyword-attached lenses (SDG, Pillar). Documents carry section-level tags against document-context lenses (Function — what kind of activity each section describes). The two-axis Map and Wedding Cake Score work by intersecting these tag sets.

What if I'm not doing sustainability?

Pick *Other* in the wizard's focus step. The project will be created empty; head to **Keywords** and **Lenses** in the left nav to build your own taxonomy, then to Settings to define a scoring rule.

Project Setup

A project bundles documents, a keyword list, lenses, and a scoring rule. This chapter covers the Setup tab, where all of that is assembled.

Setup tab

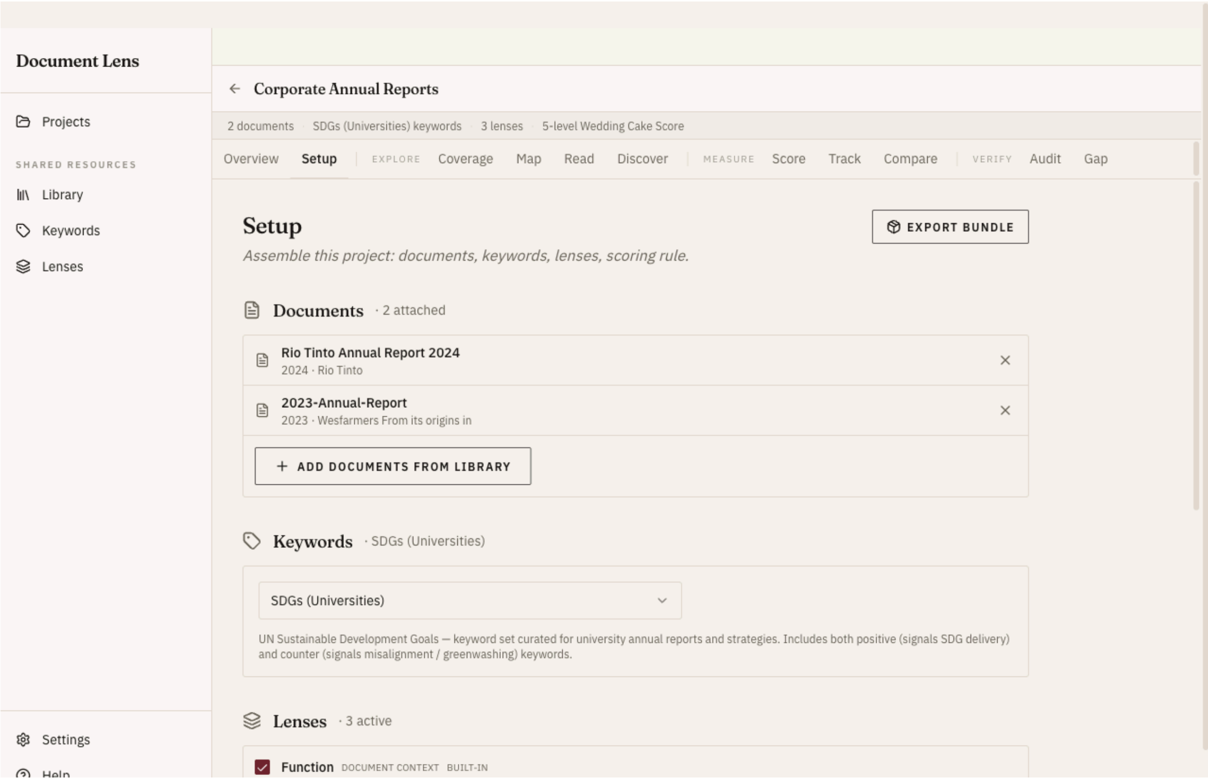

Figure 3: The Setup tab assembles documents, keyword list, lenses, and scoring rule.

Setup is where a project's components live: **documents**, **keyword list**, active **lenses**, **Function classification**, and the **scoring rule**. Every workflow tab reads from this configuration.

Documents

Add documents from Library opens a picker over your global Library — docs are stored once and can be attached to many projects. The picker has an *Import new…* button so you can import fresh files into the Library and attach them in one go.

If a document arrived via project-bundle import without its source file (see *Project bundle* topic), it appears with a yellow *Source missing* chip. Click *Locate file…* to point at the file on disk; we hash-verify the file you pick to be sure it's the same one (otherwise extracted text + sections + tags would no longer match).

Keywords + Lenses

Pick the active keyword list (one per project) and the active lenses (multiple). For sustainability projects you typically want `SDGs (Universities)` as the list and the three built-in lenses (SDG, Pillar, Function) all active.

Function classification

For Map's two-axis matrix and Score's full Wedding Cake mode, each document section needs to be tagged with a Function value. Click *Classify documents* to send each section's text to the embedding model and assign the closest Function. It's deterministic per model version. Re-running (the button becomes *Re-classify*) wipes and re-detects sections and reclassifies the *whole* corpus, not just new documents — so you only need it after adding documents or changing the Function lens.

Bundle export lives in the Setup tab header, top right. See the *Project bundle* topic.

Scoring rule

Pick which scoring rule to apply. The seeded one (5-level Wedding Cake Score) counts how many of four Functions deliver all three required Pillars. Custom rules are built in **Settings**.

Explore — See What the Corpus Contains

The Explore workflows answer "what is in these documents?" — which keywords appear where (Coverage), how topics distribute and overlap (Map), what each document actually says (Read), and what vocabulary the corpus itself uses (Discover).

Coverage

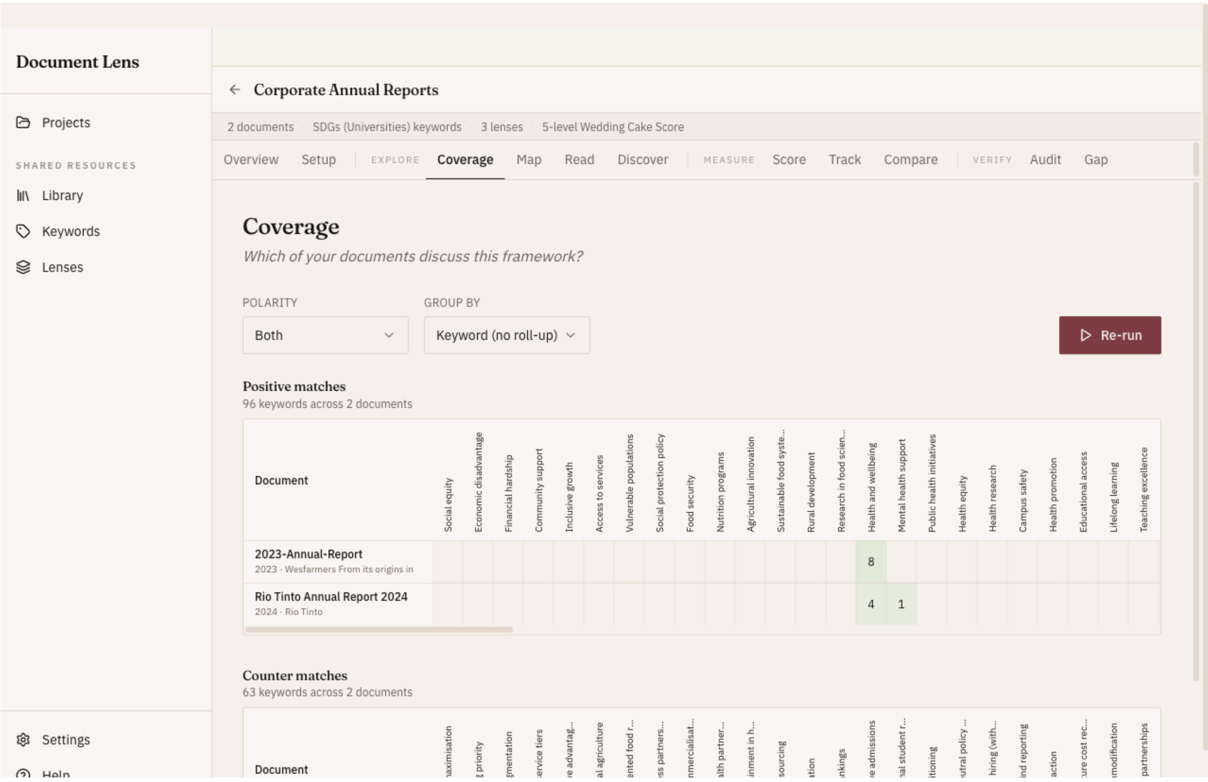

Figure 4: Coverage: documents × keywords heatmaps, positive matches above counter matches.

Which keywords appear where?

Coverage is the natural first workflow once a project is set up — the start of the *Explore* phase. It computes per-document × per-keyword match counts and displays them as a heatmap, plus a roll-up by lens value (e.g. by Pillar — Biosphere / Society / Economy) to give you the framework-level view.

Stacked positive + counter heatmaps

When the active keyword list has both polarities, Coverage shows two heatmaps one above the other (positive then counter). Honest greenwashing detection needs both — a doc with strong positive coverage but heavy counter-keyword use is the canonical "performative disclosure" pattern.

Coverage is purely local — no backend call. Click **Run coverage** (or **Re-run** after changing the polarity or group-by) and it computes in milliseconds.

Map

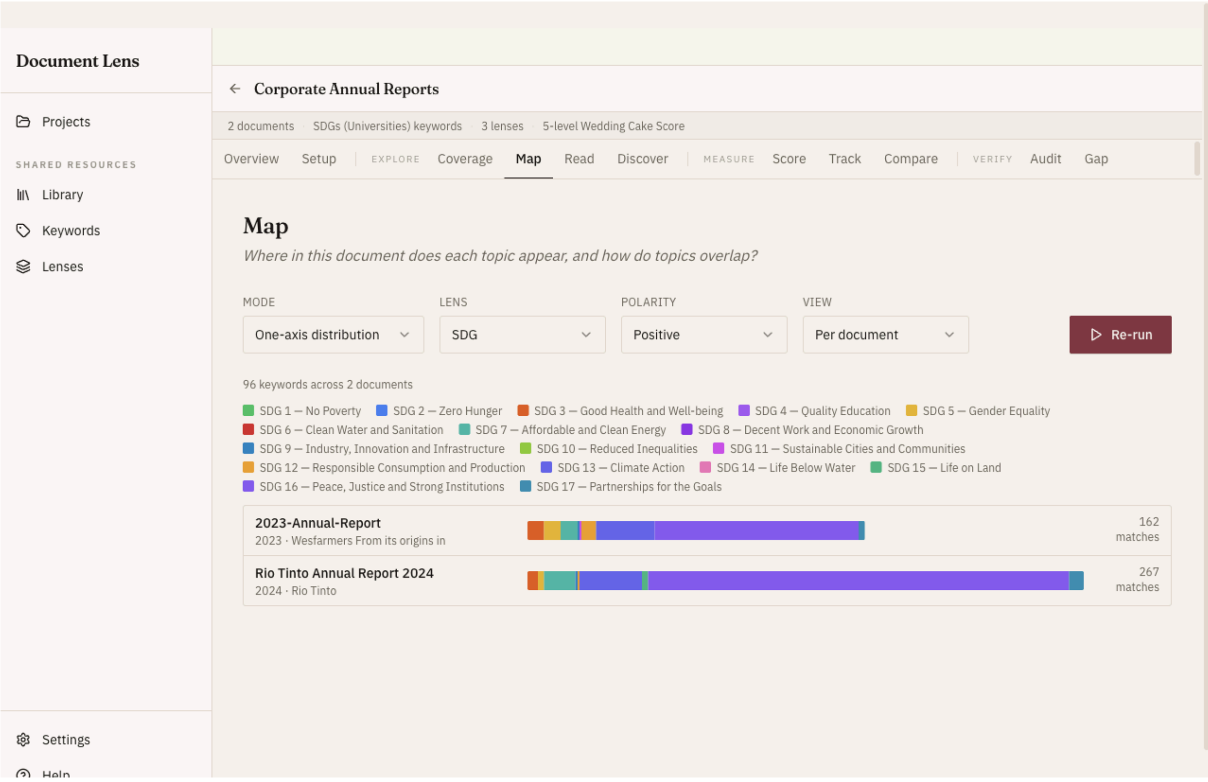

Figure 5: Map: how keyword usage distributes across lens axes.

How does each document distribute across one or two lens axes?

Map has two modes, toggled at the top:

- **One axis** (the default) — how a document's keyword usage spreads across a single keyword-attached lens (SDG, Pillar, …).
- **Two axes** — cross-tabulates a keyword-attached lens (rows) against a document-context lens (columns), e.g. the methodology's **SDG** × **Function** table. Each cell is the number of keyword hits (positive by default; counter is also selectable) in sections classified as that column value, for keywords tagged with that row value. The Rows / Columns dropdowns default to the first eligible lens of each kind.

Use the two-axis view to test the Wedding Cake hypothesis per document: does this report deliver SDG 13 (Climate Action) through Operations, or only through Awareness? An empty Function column means the doc isn't delivering any SDG via that activity type.

Two-axis mode needs a document-context lens classified on Setup. With no classified documents the matrix still renders but every cell is zero; a partly-classified corpus shows a "classification incomplete" banner with a Classify-now shortcut.

Read

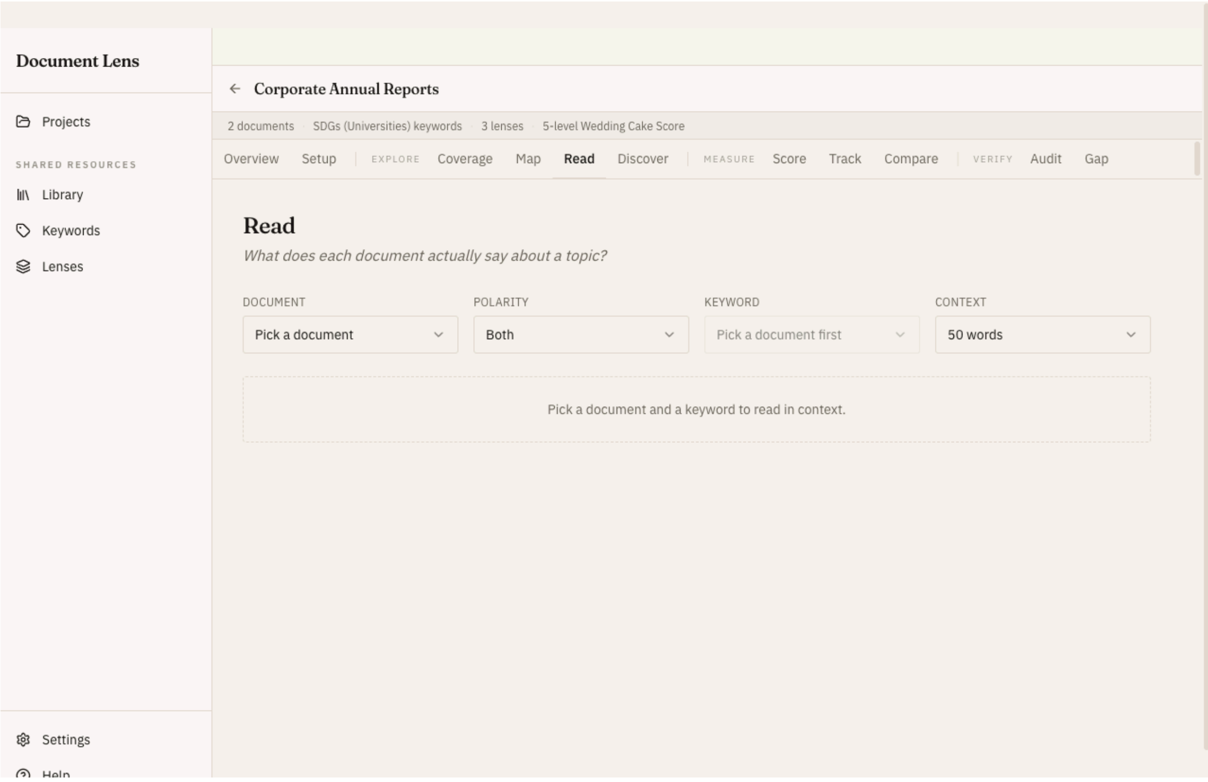

Figure 6: Read: each keyword match in its surrounding context.

What does each document actually say about a topic?

Read is the concordance view: pick a document and a keyword, see every match with surrounding context (50 / 100 / 250 words). The keyword dropdown is filtered to ones that actually appear in the picked doc, so you don't pick a keyword and see "no matches".

Locating the passage in the source PDF

Several navigation aids, in increasing reliability. **Copy phrase** is on every match card; the two PDF aids appear only when the source is a PDF present on disk; **Open source file** is a single button in the results header (not per-match):

- **Preview** (PDF on disk only) — opens the source PDF in an in-app modal at the right page (Chromium's built-in PDFium viewer). Hit ⌘F and paste the keyword.
- **Open at page N** (PDF on disk only) — opens in your OS PDF viewer at the right page via `file://…#page=N`. Preview / Acrobat honour the fragment; viewers that don't, gracefully open at page 1.
- **Open source file** (header button) — opens the document at page 1 in your OS viewer.
- **Copy phrase** — three-word snippet on your clipboard. Paste into any viewer's Find. Three words is short enough to dodge the footnote / header noise that the extractor sometimes inlines.

Section labels

When a match is in a classified document, the card shows the containing section as `§ N/M` plus the first ~80 chars of that section's text — usually the heading, so you can tell what part of the document you're looking at without opening the PDF.

Discover

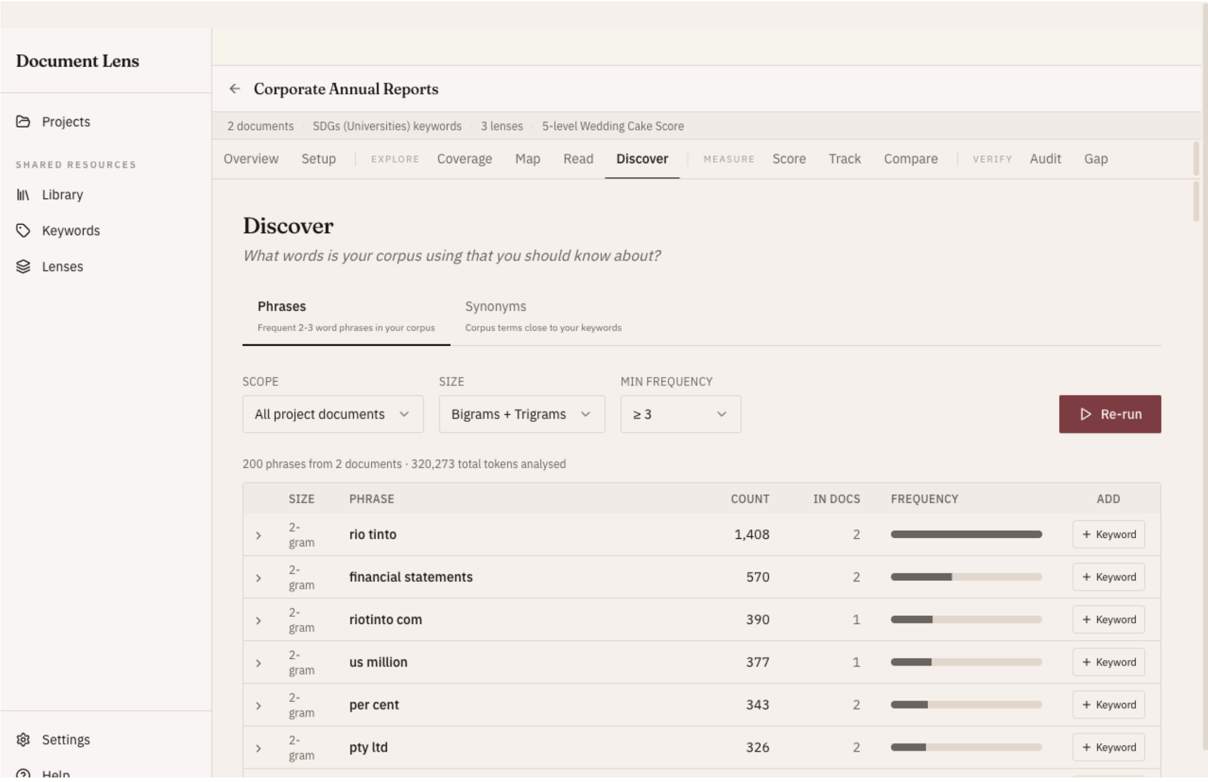

Figure 7: Discover: phrases and synonyms the corpus uses that your list may be missing.

What phrases / synonyms is the corpus using that you should know about?

Phrases (n-grams)

Extracts frequent 2- and 3-word phrases from the corpus, ranked by count (with an "in docs" document-spread column shown alongside). Useful for finding domain-specific terminology you might want to add as keywords. The *Keyword* button on each row adds the phrase straight to the active list as a positive keyword.

Synonyms

For each enabled keyword, ranks corpus n-grams by **semantic similarity** (sentence-embedding cosine — meaning, not letters). High similarity = the candidate is likely a synonym worth tracking. Two ways to keep one:

- **Synonym** — attaches the candidate as a synonym of the parent keyword (preserves provenance, counts in Coverage automatically).
- **Keyword** — adds it as a first-class entry on the list with the same polarity. Useful for counter-keyword discovery — promote a candidate counter-narrative term to its own keyword rather than burying it as a synonym.

If Synonyms returns "no candidates above the similarity threshold", drop *Min similarity* to 0.3 and bump *Top per keyword* to 12+. Small corpora (a few documents) genuinely have less to surface.

Measure — Put Numbers on It

The Measure workflows quantify the corpus: rate each document on a rubric (Score), follow a metric across years (Track), and rank documents (Compare).

Score

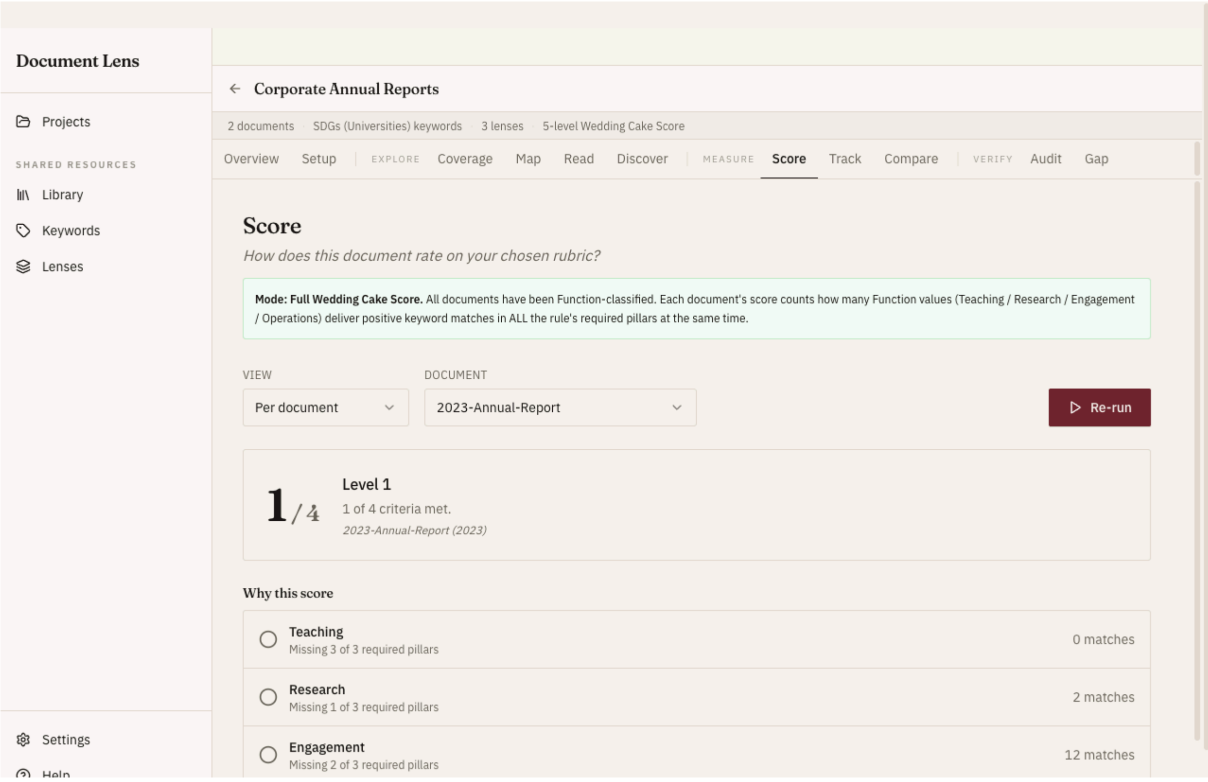

Figure 8: Score: each document rated on the chosen rubric, with the why.

How does each document rate on your chosen rubric?

Score runs each document through the project's **scoring rule** and gives it a level — a small integer like `3 / 4`. The default rule is the **5-level Wedding Cake Score** (levels 0–4). The big number is "how many criteria this document met"; the **Why this score** panel underneath lists exactly which ones.

Check the banner first — it changes what the number means

The coloured banner at the top tells you which mode you're in, and the same `3 / 4` means different things in each:

- **Full Wedding Cake Score** (green) — every document has been Function-classified. The score counts how many of the four **Functions** (Teaching / Research / Engagement / Operations) deliver positive keyword matches in *all* the rule's required **Pillars** at once. The default rule requires three — Biosphere, Society, Economy (Partnership is the connector, not a required layer). So `3 / 4` = three functions cover all three pillars.

- **v1 Pillar coverage prerequisite** (yellow) — documents aren't Function-classified yet, so the full score can't be computed. This shows a fallback proxy: how many of the required Pillars the document mentions positively. Run **Function classification** on the Setup tab (the banner has a "Classify now" button) to upgrade to the full score.

Why this score

Below the big number is the breakdown — one row per criterion (a Function in full mode, a Pillar in v1 mode). A green check means the criterion was **met**, a hollow circle means **unmet**, and the right-hand count is how many keyword matches contributed. A row reads like *"Teaching — Delivers all required pillars"* (met) or *"Research — Missing 1 of 3 required pillars"* (unmet). This is where you see *why* a document landed at its level.

Per document vs Project distribution

The **View** switcher toggles between a single document (the big score card + its Why-this-score breakdown) and the **Project distribution** — a histogram of how many documents landed at each level (0…max), with counts and percentages. Use the distribution to see how the whole corpus stacks up and to spot outliers.

Score uses your positive keywords (it measures delivery), the rule from Setup, and — for full mode — the Function classification. Change any of those and hit **Re-run**.

Track

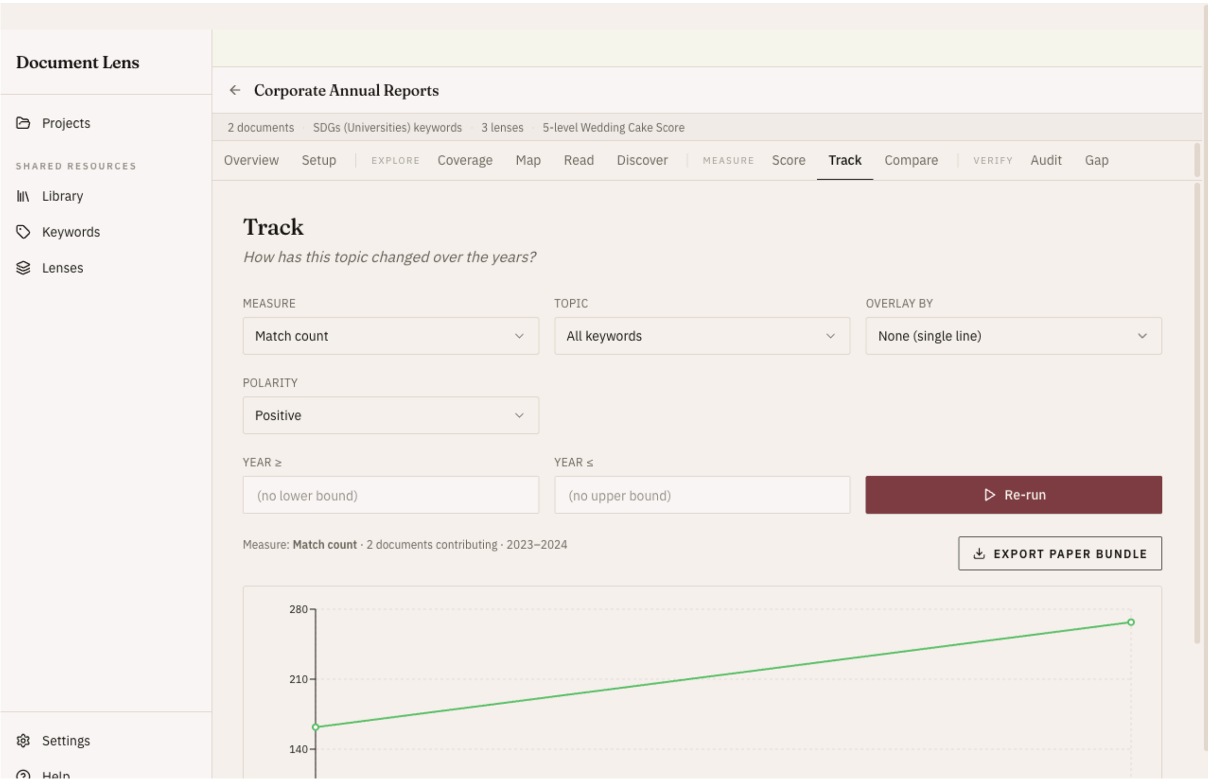

Figure 9: Track: the metric across years.

How does the metric move year-over-year?

Track aggregates a chosen **measure** across the corpus by year — useful for "has disclosure improved over the last decade?" questions. The three measures are **Match count**, **Coverage %** (share of that year's documents with at least one match), and **Score** (the scoring rule's per-year average). It needs at least two distinct years to plot.

Overlay by company / sector

The default series is one line for the whole corpus. Switching *Overlay by* to Company or Sector splits it — one palette-coloured trend per company, so you can compare "is BHP improving faster than Rio?" directly.

Polarity

The Polarity selector sets which keywords feed the chart (positive or counter). To see the greenwashing trend directly, set *Overlay by* to **Polarity** — that draws positive and counter as two coloured lines on one chart (the Polarity selector is hidden in this mode, since both are already shown).

The *Export paper bundle* button on Track ships your current chart as a PNG plus the underlying data — drop straight into a paper. See the *Paper-ready bundle* topic.

Compare

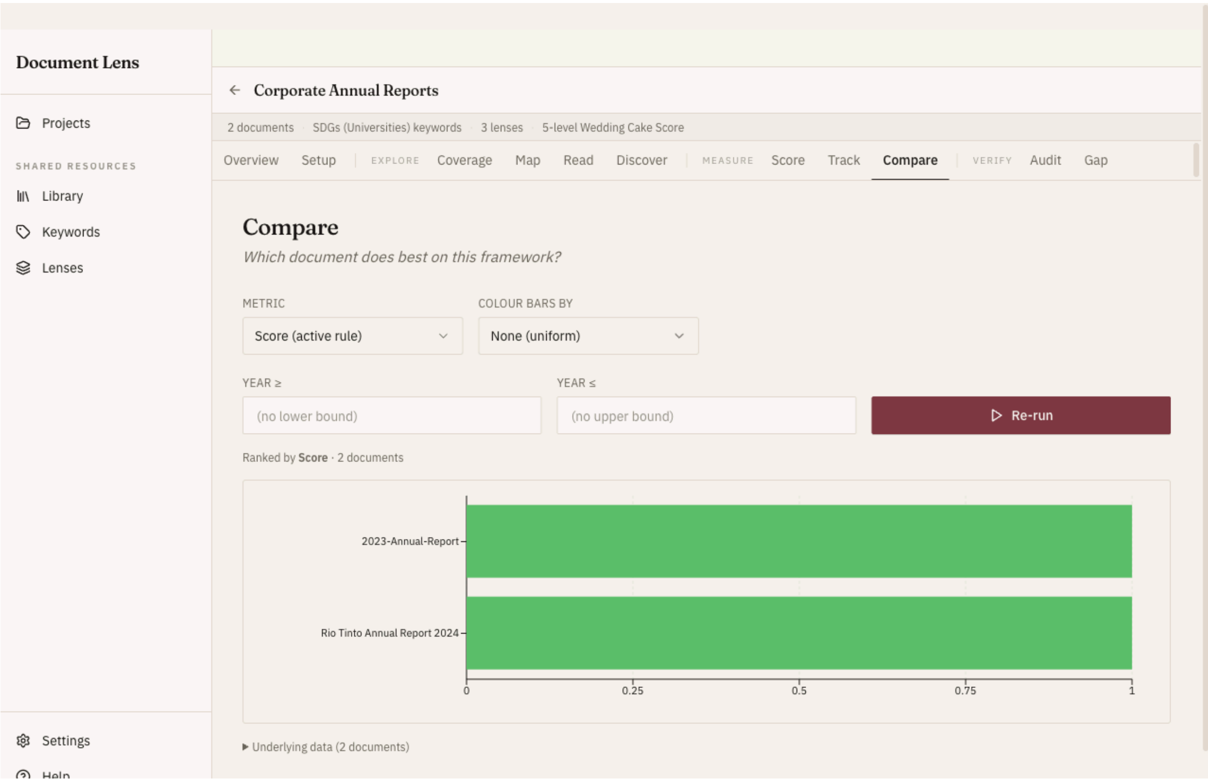

Figure 10: Compare: documents ranked on the chosen metric.

Rank documents on the chosen metric.

Compare is "Track without the time dimension" — same per-document measures, ranked by value rather than aggregated by year. Headline use case: "which report does best on the Wedding Cake Score?"

Per-keyword filter

For match-count and distinct-keywords metrics, the Keyword selector narrows the ranking to a single keyword. Lets you ask “which doc talks most about *circular economy*?” directly.

Filters + colour-by

A year range (min / max) plus company and sector multi-select filters narrow the ranked set without changing the metric. Colour bars by Company / Year / Sector to spot clusters visually.

Verify — Check the Evidence Holds Up

The Verify workflows stress-test your findings: are keywords used in the right context (Audit), and where does tone run ahead of substance (Gap)?

Audit

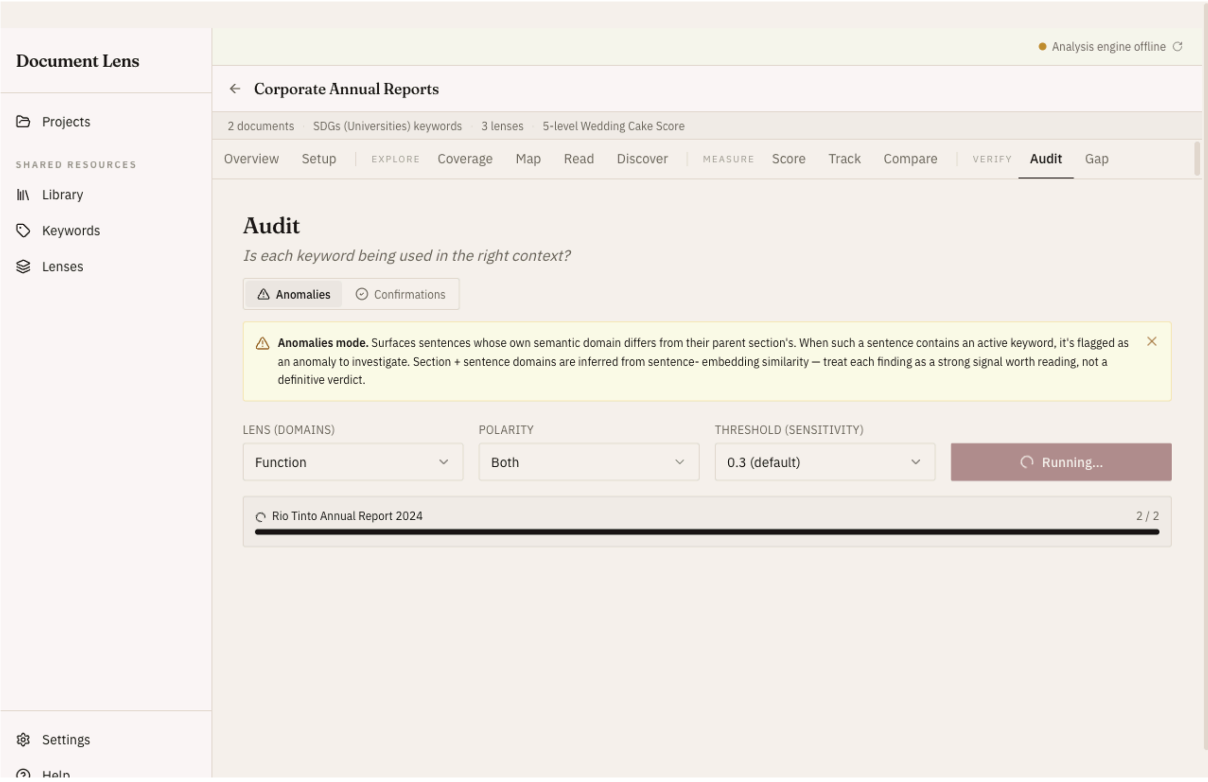

Figure 11: Audit: is each keyword used in the right context?

Is each keyword being used in the right context?

The methodology calls this the "contextual relevance check": if a keyword appears in a section whose semantic domain doesn't match the keyword's intended Function, that's a signal worth investigating.

Anomalies

Sends each document to the backend's `/semantic/structural-mismatch` endpoint, which classifies every sentence + every parent section and flags sentences whose own domain disagrees with their parent's. We surface only the keyword-bearing flagged sentences. Use the threshold to trade noise for sensitivity.

Per-doc responses are cached by content hash + lens config + threshold. The first run on a project is slow (sentence embedding is expensive); re-runs with the same inputs are instant.

Confirmations

The inverse view: keyword usages that *did* land in their expected-Function section. Defensible "yes, this is being used in the right context" evidence for a sceptical reviewer. No backend call — uses the cached Function classifications from Setup directly. Confidence (how strongly the section matched the Function) drives the severity bucket and is shown as a percentage on the badge (e.g. `medium · 38%`).

Gap

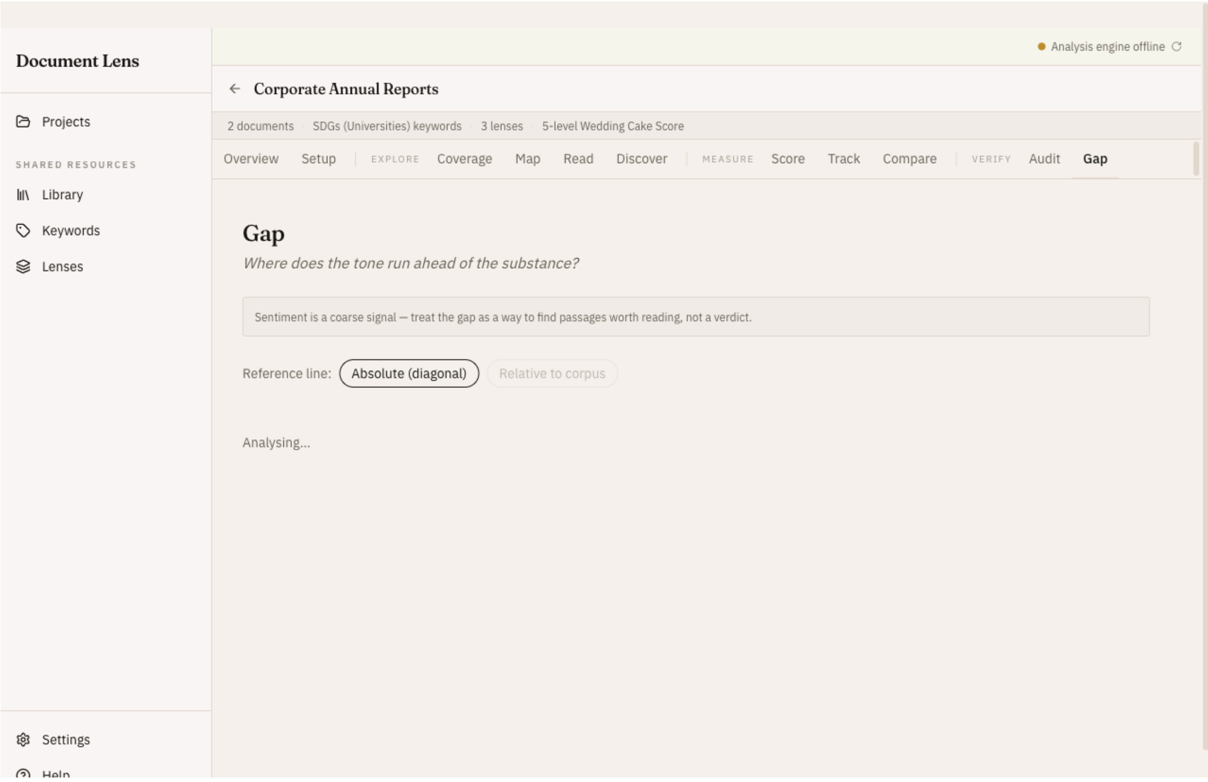

Figure 12: Gap: sentiment against keyword substance — tone running ahead of delivery.

Where does the tone run ahead of the substance?

Gap plots each document, section, and keyword in **tone (sentiment)** × **substance (keyword polarity)** space. Distance above the 1:1 diagonal is the "talk exceeds walk" measure — the gap between what is said and what is delivered.

The gap is more informative than either axis alone, for a specific reason: in corporate disclosure the tone is uniformly positive, so absolute sentiment barely varies. What varies meaningfully is how far the tone runs ahead of (or behind) the substance. Subtracting one axis from the other cancels out the baseline positivity that makes raw sentiment useless here. You're left with a residual: tone not justified by substance.

Quadrant guide

	High substance (delivery)	Low substance (counter / sparse)
High tone	aligned — genuine good news	talk > walk ⟶ performative / greenwashing
Low tone	walk > talk ⟶ understated / candid	aligned — honest about gaps

Polarity vs sentiment

These two axes are orthogonal and must not be conflated:

- **Polarity** is a property of the *keyword* — curated by the researcher: positive keywords signal delivery, counter keywords signal performative vocabulary.
- **Sentiment** is a property of the *text* — inferred by the model from the tone of the surrounding prose.

Greenwashing is positive in tone but hollow in substance, so the gap between them is the signal.

Gap is a coarse signal for surfacing passages that warrant closer reading — not a verdict on a document's integrity. Use **Read** to inspect the flagged passages in context.

Sharing & Export

Paper-ready bundle

A ZIP designed to drop straight into an academic paper. Captures the **current Track view**, not the whole project.

What's in it

- `methodology.md` — auto-generated configuration blurb (measure, polarity, filters, scoring rule)
- `data.csv` — the pivoted year × series matrix
- `documents.csv` — the contributing documents with title / filename / year / company / sector
- `chart.png` — the chart you're looking at, rendered at 2× for retina-ish quality (only when a chart is currently rendered)
- `per-document.csv` — per-doc scores (only when measure = score and there are per-document scores)

Saved via a native file dialog. No backend round-trip — composed client-side from the Track result + a snapshot of the rendered chart SVG.

This is different from the *project bundle* (.lens). Paper-ready is for publication; project bundle is for sharing the whole project with a colleague.

Project bundle (.lens)

A `.lens` file containing the full project state — for researcher-to-researcher sharing. Local-first, single-user, no auth or central DB required: export, email, the recipient imports.

Export

Setup tab ⟶ top-right *Export bundle* button. Writes the project's documents (metadata + extracted text + per-page text + sections + tags), keyword list, lenses, scoring rule, and (by default) the original source files into a single ZIP.

Import

Projects page ⟶ *Import bundle* button (or the same option on the empty state). Shows a preview first: counts of what will be created vs reused, plus warnings (e.g. when the bundle includes no source files). Click *Import as new project* to apply.

How identity is handled

- **Documents** are matched by file content hash. Same PDF already in your Library ⟶ the existing doc is reused. New PDF + bundle includes the file ⟶ written to your app data folder. New PDF

+ no file in the bundle ⟶ metadata-only doc, marked *Source missing* in Setup until you locate the file.

- **Built-in lenses** are matched by name (so SDG / Pillar / Function and any other built-in are reused, not duplicated), the keyword list is matched by its *source* identifier, and a built-in scoring rule (e.g. the Wedding Cake Score) is matched by name — your sustainability defaults are never duplicated by an import.
- **Custom lenses, lists, and rules** get fresh IDs. Name collisions get a `(imported)` suffix; you can rename anytime.
- **Project name** collisions also get the suffix. Two researchers can each have their own "Acme 2024 Sustainability" without overwriting.

Re-linking missing source files

When a bundle arrives without source files, the docs analyse fine (Coverage, Score, Track, Read concordance, Audit Confirmations all work from cached text), but Preview / Open in viewer is unavailable. Setup shows a yellow *Source missing* chip on each affected doc; click *Locate file…* and pick the file on disk. We hash-verify it matches before re-linking, so you can't accidentally relink the wrong file and silently corrupt the analysis.